

3 The photograph shows a sea slug of the genus *Aplysia*, a marine animal.

external
gills



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Magnification $\times 1.0$

Aplysia can be kept in tanks in a laboratory.

Aplysia absorb oxygen using external gills.

When the gills are touched, the gills are completely withdrawn into the body by a reflex action.

The gills re-emerge after some time.

The student formed the following hypothesis:

The more frequently the gills are touched the less time it takes them to re-emerge.

Plan an investigation to find evidence to support or reject this hypothesis.

(a) Describe preliminary practical work that you might undertake to ensure your proposed method would provide quantitative results.

(3)

(b) Devise a detailed method, including an explanation of how you would control and monitor important variables.

(10)

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(c) Describe how your results should be recorded, presented and analysed, in order to draw conclusions from your investigation.

(4)

[illegible]

(d) Suggest **four** limitations of your proposed method.

(4)

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(Total for Question 3 = 21 marks)

TOTAL FOR PAPER = 50 MARKS